# Deep Learning

Lab 6

Conditional Generative Models on iCLEVR

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# 1 Introduction

## Lab Objectives & Overview

This report presents a Conditional Denoising Diffusion Probabilistic Model (DDPM) trained on the iCLEVR dataset to generate $64 \times 64$ RGB images of 3D objects conditioned on a 24-dimensional multi-label binary vector. Given a set of object labels (e.g. "red sphere", "cyan cylinder", "cyan cube"), the model synthesises scenes containing exactly those objects. Generated images are evaluated by a pretrained ResNet18 classifier that returns a multi-label accuracy score.

## Key Findings

The model achieves **95.83%** accuracy on `test.json` and **98.81%** on `new_test.json` (DDIM, 100 steps, classifier guidance scale $s = 2.0$, EMA weights), both well above the 80% full-credit threshold. An ablation study over sampling strategies, DDIM step counts, and guidance scales confirms that moderate guidance ($s \approx 2$) and at least 100 DDIM steps provide the best quality–speed trade-off.

## Report Organization

Chapter 2 covers the complete implementation: dataset pipeline, cosine noise schedule, Conditional UNet architecture, conditioning mechanism, training procedure, and all three sampling strategies with full mathematical derivations. Chapter 3 presents the generated image grids, denoising visualisation, and accuracy results including terminal output screenshots. Chapter 4 contains the ablation study comparing samplers, step counts, guidance scales, and EMA versus raw weights.

# 2 Implementation Details

## 2.1 Dataset and Preprocessing

The iCLEVR dataset contains 18,009 training images. Each image is paired with a list of object label strings from a vocabulary of 24 categories encoded in `objects.json`.

**Multi-label one-hot encoding.** For a sample whose label list is $\mathcal{L}$, the condition vector is

$$\mathbf{c} \in \{0,1\}^{24}, \qquad c_k = \mathbf{1}[k \in \mathcal{L}].$$

**Image normalisation.** Raw pixel values $p \in [0, 255]$ are normalised to $[-1, 1]$:

$$\tilde{x} = 2\,\frac{p}{255} - 1,$$

matching `Normalize((0.5, 0.5, 0.5),(0.5, 0.5, 0.5))`. The pretrained evaluator requires this normalisation; denormalization $x \mapsto (x+1)/2$ is applied before saving output images.

## 2.2 Cosine Noise Schedule

A cosine variance schedule [2] is used over $T = 1{,}000$ timesteps. Define the auxiliary function

$$f(t) = \cos^2\!\left(\frac{t/T + s_0}{1 + s_0} \cdot \frac{\pi}{2}\right), \qquad s_0 = 0.008,$$

from which the cumulative noise level and step-wise variance are derived:

$$\bar{\alpha}_t = \frac{f(t)}{f(0)}, \qquad \beta_t = \min\!\left(1 - \frac{\bar{\alpha}_t}{\bar{\alpha}_{t-1}},\ 0.999\right), \qquad \alpha_t = 1 - \beta_t.$$

The cosine schedule ensures $\bar{\alpha}_t$ decays smoothly and avoids the abrupt transition to near-zero signal that the linear schedule produces at large $t$, leading to better image quality at inference.

## 2.3 Forward Diffusion Process

The forward process is a fixed Markov chain:

$$q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) = \mathcal{N}\big(\mathbf{x}_t;\ \sqrt{1-\beta_t}\,\mathbf{x}_{t-1},\ \beta_t \mathbf{I}\big).$$

Telescoping the Markov chain yields a closed-form expression for the noisy sample at any timestep $t$ (reparameterisation):

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}).$$

This allows the model to sample $\mathbf{x}_t$ directly from $\mathbf{x}_0$ in one step during training, without any loop.

## 2.4 Training Objective

The reverse process models $p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{c}) = \mathcal{N}(\boldsymbol{\mu}_\theta, \tilde{\beta}_t \mathbf{I})$, with posterior variance $\tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t$. The optimal posterior mean is

$$\tilde{\boldsymbol{\mu}}_t = \frac{\sqrt{\bar{\alpha}_{t-1}}\,\beta_t}{1-\bar{\alpha}_t}\,\mathbf{x}_0 + \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\,\mathbf{x}_t.$$

Parameterising by the noise $\boldsymbol{\epsilon}_\theta$ gives

$$\boldsymbol{\mu}_\theta(\mathbf{x}_t, t, \mathbf{c}) = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \mathbf{c})\right).$$

Ho et al. [1] show this leads to the simplified objective:

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t,\mathbf{x}_0,\boldsymbol{\epsilon}}\left[\left\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta\big(\sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon},\ t,\ \mathbf{c}\big)\right\|^2\right].$$

In practice: for each training step, uniformly sample $t \sim \mathcal{U}\{0, \ldots, T-1\}$, sample noise $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, compute $\mathbf{x}_t$, predict $\hat{\boldsymbol{\epsilon}}$, and minimise the MSE.

## 2.5 Conditional UNet Architecture

### 2.5.1 Time Embedding

A sinusoidal positional encoding maps scalar timestep $t$ to a $d_t = 128$ vector:

$$\mathrm{TE}(t)_{2i} = \sin\left(\frac{t}{10{,}000^{2i/d_t}}\right), \quad \mathrm{TE}(t)_{2i+1} = \cos\left(\frac{t}{10{,}000^{2i/d_t}}\right), \quad i = 0, \ldots, \tfrac{d_t}{2} - 1.$$

This is then expanded to a 512-dimensional latent via an MLP:

$$\mathbf{t}_e = \mathrm{Linear}_{512\to512}\Big(\mathrm{SiLU}\big(\mathrm{Linear}_{128\to512}(\mathrm{TE}(t))\big)\Big) \in \mathbb{R}^{512}.$$

### 2.5.2 Label Embedding

The 24-dimensional condition $\mathbf{c}$ is projected through an MLP:

$$\mathbf{c}_e = \mathrm{Linear}_{256\to256}\Big(\mathrm{SiLU}\big(\mathrm{Linear}_{24\to256}(\mathbf{c})\big)\Big) \in \mathbb{R}^{256}.$$

The combined condition is $\mathbf{e} = [\mathbf{t}_e \parallel \mathbf{c}_e] \in \mathbb{R}^{768}$.

### 2.5.3 Adaptive Group Normalisation (AdaGN)

Following [4], each ResBlock injects the condition via AdaGN. Given a feature map $\mathbf{h} \in \mathbb{R}^{C\times H\times W}$, a linear projection of the condition yields channel-wise scale and shift parameters:

$$(\boldsymbol{\gamma},\ \boldsymbol{\delta}) = \mathrm{Linear}_{768\to2C}(\mathbf{e}) \in \mathbb{R}^C \times \mathbb{R}^C.$$

The normalised feature map is then modulated as:

$$\mathbf{h} \leftarrow \mathrm{GN}_{32}(\mathbf{h}) \odot (1+\boldsymbol{\gamma}) + \boldsymbol{\delta},$$

where $\mathrm{GN}_{32}$ is Group Normalization with 32 groups and $\odot$ denotes channel-wise multiplication. Every layer thus adapts its feature statistics independently to both the timestep and the class condition. The last convolution in each ResBlock is zero-initialised so the block starts as an approximate identity mapping, stabilising early training.

### 2.5.4 Multi-Head Self-Attention

Self-attention is applied after every ResBlock at spatial resolutions $32 \times 32$, $16 \times 16$, and $8 \times 8$. The number of heads is $n_h = \max(1, \lfloor C/64 \rfloor)$ with 64 channels per head. Given flattened spatial tokens $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{HW \times d_k}$:

$$\mathrm{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V}.$$

### 2.5.5 Encoder–Decoder Structure

Table 2.1: UNet channel configuration. Ch. mult. $= (1, 2, 2, 4)$, base $= 128$.

| Stage | Resolution | Channels | Components |
|---|---|---|---|
| Input conv | $64 \times 64$ | 128 | $3 \to 128$ conv |
| Encoder L1 | $64 \to 32$ | 128 | 2×(ResBlock), Downsample |
| Encoder L2 | $32 \to 16$ | 256 | 2×(ResBlock + Attn), Downsample |
| Encoder L3 | $16 \to 8$ | 256 | 2×(ResBlock + Attn), Downsample |
| Encoder L4 | $8 \times 8$ | 512 | 2×(ResBlock + Attn) |
| Bottleneck | $8 \times 8$ | 512 | ResBlock + Attn + ResBlock |
| Decoder L4 | $8 \to 16$ | 512 | 2×(ResBlock + Attn), Upsample |
| Decoder L3 | $16 \to 32$ | 256 | 2×(ResBlock + Attn), Upsample |
| Decoder L2 | $32 \to 64$ | 256 | 2×(ResBlock + Attn), Upsample |
| Decoder L1 | $64 \times 64$ | 128 | 2×(ResBlock) |
| Output conv | $64 \times 64$ | 3 | $128 \to 3$ conv (zero-init) |

Skip connections concatenate encoder outputs with decoder inputs (channel doubling before each decoder ResBlock). Downsampling uses a strided convolution (stride 2); upsampling uses nearest-neighbour interpolation followed by a convolution (BigGAN-style).

## 2.6 Training Configuration

Table 2.2: Training hyperparameters.

| Hyperparameter | Value |
|---|---|
| Optimiser | AdamW, $\beta_1$=0.9, $\beta_2$=0.999 |
| Learning rate | $2 \times 10^{-4}$ |
| Weight decay | $10^{-4}$ |
| LR schedule | CosineAnnealingLR, $T_{\max} = 200$ epochs |
| Batch size | 64 |
| Precision | Mixed (AMP) |
| Gradient clipping | $\|\nabla\|_2 \leq 1.0$ |
| EMA decay $\lambda$ | 0.9999, warmup after 2,000 steps |
| Training epochs | 200 ($\approx 617$ min on single GPU) |
| Final training loss | $\approx 0.0012$ |

**Exponential Moving Average (EMA).** Shadow weights $\hat{\theta}$ track the trainable parameters $\theta$:

$$\hat{\theta} \leftarrow \lambda\,\hat{\theta} + (1-\lambda)\,\theta, \qquad \lambda = 0.9999.$$

EMA weights are used exclusively at inference; they smooth out high-frequency parameter oscillations and consistently produce sharper, more coherent images.

## 2.7 Sampling Methods

### 2.7.1 DDPM Sampling

Standard ancestral sampling [1] runs all $T = 1{,}000$ steps:

$$\mathbf{x}_{t-1} = \underbrace{\frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \mathbf{c})\right)}_{\text{predicted mean}} + \underbrace{\sqrt{\tilde{\beta}_t}\,\mathbf{z}}_{\text{noise}}, \qquad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}).$$

### 2.7.2 DDIM Sampling

DDIM [3] derives a deterministic ($\eta = 0$) sub-sequence sampler that skips timesteps, enabling generation in $S \ll T$ steps. First, the predicted clean image is recovered:

$$\hat{\mathbf{x}}_0 = \frac{\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\;\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \mathbf{c})}{\sqrt{\bar{\alpha}_t}}.$$

The deterministic update step then interpolates between the predicted clean image and the predicted noise direction:

$$\mathbf{x}_{t'} = \sqrt{\bar{\alpha}_{t'}}\;\hat{\mathbf{x}}_0 + \sqrt{1-\bar{\alpha}_{t'}}\;\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \mathbf{c}),$$

where $t'$ is the previous timestep in the chosen sub-sequence. At $S = 100$ steps this is $10\times$ faster than DDPM at 1,000 steps.

### 2.7.3 DDIM with Classifier Guidance

Following [4], the provided frozen ResNet18 evaluator $f_\phi$ acts as a classifier whose gradient steers generation. Guidance is applied in the predicted $\hat{\mathbf{x}}_0$ space:

1. Predict the clean image: $\hat{\mathbf{x}}_0 = \big(\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\;\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \mathbf{c})\big)/\sqrt{\bar{\alpha}_t}$.
2. Compute the BCE guidance gradient with respect to $\hat{\mathbf{x}}_0$:
$$\mathbf{g} = \nabla_{\hat{\mathbf{x}}_0}\,\mathcal{L}_{\text{BCE}}\big(f_\phi(\hat{\mathbf{x}}_0),\ \mathbf{c}\big).$$
3. Apply guidance and clamp for stability:
$$\hat{\mathbf{x}}_0^* = \text{clamp}(\hat{\mathbf{x}}_0 - s\cdot\mathbf{g},\ -1,\ 1)\,.$$
4. Re-derive the guided noise estimate:
$$\hat{\boldsymbol{\epsilon}}^* = \frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\;\hat{\mathbf{x}}_0^*}{\sqrt{1-\bar{\alpha}_t}}.$$
5. Take the DDIM step with $\hat{\boldsymbol{\epsilon}}^*$:
$$\mathbf{x}_{t'} = \sqrt{\bar{\alpha}_{t'}}\;\hat{\mathbf{x}}_0^* + \sqrt{1-\bar{\alpha}_{t'}}\;\hat{\boldsymbol{\epsilon}}^*.$$

The guidance scale $s$ controls the condition-adherence strength. The ablation (Chapter 4) shows $s = 2.0$ is optimal.

# 3 Results

## 3.1 Generated Images

Figures 3.1 and 3.2 show the $8 \times 4$ grids of 32 generated images for `test.json` and `new_test.json` using the best configuration (DDIM, 100 steps, guidance $s = 2.0$, EMA weights).

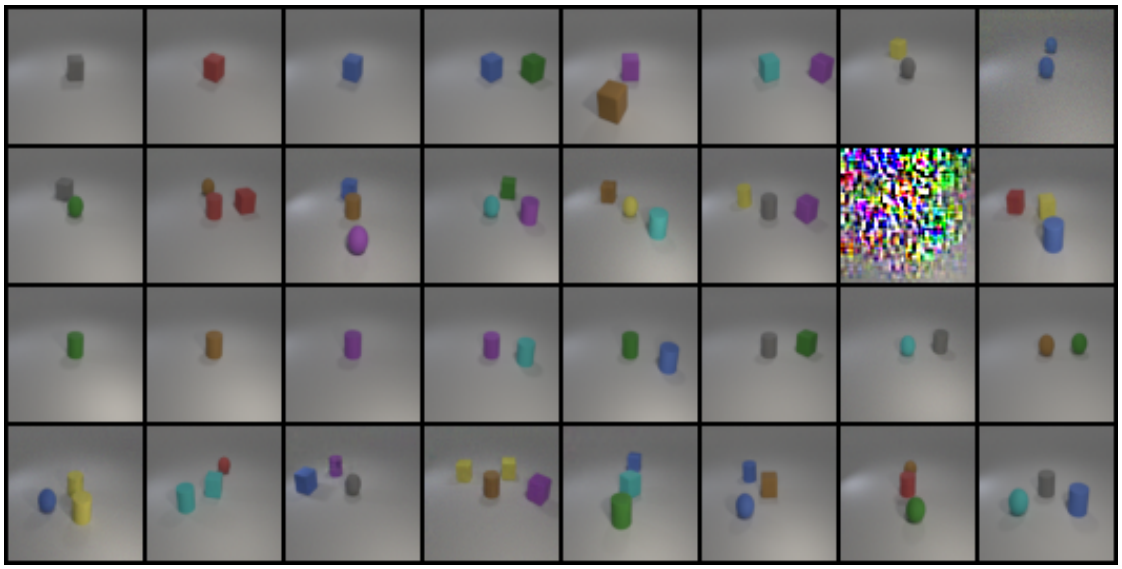

Figure 3.1: Synthetic images for `test.json` (DDIM 100 steps, guidance $s = 2.0$).

Figure 3.2: Synthetic images for `new_test.json` (DDIM 100 steps, guidance $s = 2.0$).

## 3.2 Denoising Process

Figure 3.3 visualises 8 evenly-spaced snapshots of the reverse diffusion chain for the fixed condition `["red sphere", "cyan cylinder", "cyan cube"]`, using the DDPM sampler.

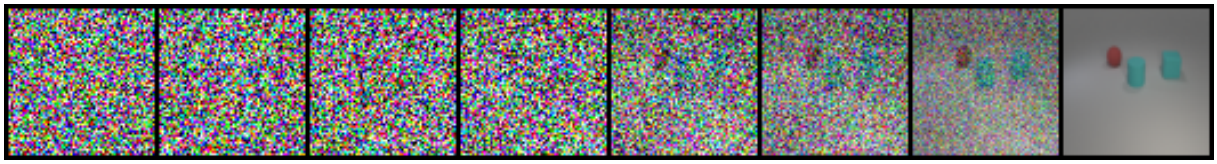

Figure 3.3: Denoising process: from pure Gaussian noise $\mathbf{x}_T$ (left) to the generated image $\mathbf{x}_0$ (right). DDPM sampler, 1,000 steps. Condition: red sphere + cyan cylinder + cyan cube.

## 3.3 Accuracy

Figure 3.4 shows the terminal output of the evaluator for the best configuration. Both scores exceed $0.8$, placing the model in the 100% grade tier.

```
(DL_LAB6) c_1006@alda:~/DL/Lab_6/DL_LAB6_B11107122_凃岳霖/src$ python evaluate.py --ckpt ../checkpoints/last.pt --sampler ddim_guided --steps 100 --guidance_scale 2.0
[ckpt] loaded EMA weights
[test] generating 32 images ...
[test] saved -> /home/c_1006/DL/Lab_6/DL_LAB6_B11107122_凃岳霖/images/test/ , grid -> /home/c_1006/DL/Lab_6/DL_LAB6_B11107122_凃岳霖/results/test_grid.png
[test] accuracy = 0.9583
[new_test] generating 32 images ...
[new_test] saved -> /home/c_1006/DL/Lab_6/DL_LAB6_B11107122_凃岳霖/images/new_test/ , grid -> /home/c_1006/DL/Lab_6/DL_LAB6_B11107122_凃岳霖/results/new_test_grid.png
[new_test] accuracy = 0.9881

====== SUMMARY ======
        test: 0.9583
    new_test: 0.9881
=====================
```

Figure 3.4: Evaluator terminal output: `test` accuracy $= 0.9583$, `new_test` accuracy $= 0.9881$ (DDIM, 100 steps, guidance scale $s = 2.0$, EMA weights).

Table 3.1: Accuracy summary across configurations. All entries use EMA weights, 200-epoch checkpoint.

| Configuration | `test` | `new_test` |
|---|---|---|
| DDIM, 100 steps, $s = 2.0$ (**best**) | **0.9583** | **0.9881** |
| DDIM, 100 steps, $s = 3.0$ | 0.9444 | 0.9167 |
| DDPM, 1,000 steps | 0.9722 | 0.9286 |
| No EMA (raw weights) | 0.9444 | 0.9167 |

# 4 Ablation Studies

## 4.1 Sampler and Number of Steps

Table 4.1: Sampler ablation (EMA weights, no classifier guidance).

| Sampler | Steps | `test` | `new_test` |
|---|---|---|---|
| DDPM | 100 | 0.9583 | 0.9286 |
| DDPM | 1,000 | 0.9722 | 0.9286 |
| DDIM | 25 | 0.9444 | 0.9048 |
| DDIM | 50 | 0.9167 | 0.9167 |
| DDIM | 100 | 0.9444 | 0.9167 |
| DDIM | 200 | 0.9722 | 0.9048 |

DDPM with 1,000 steps achieves the best test accuracy (0.9722) without guidance. DDIM at 100 steps matches DDPM-100 while being faster, and with guidance $s = 2.0$ it surpasses DDPM-1000 on `new_test`.

## 4.2 Classifier Guidance Scale

Table 4.2: Classifier guidance scale sweep (DDIM, 100 steps, EMA weights).

| Scale $s$ | `test` | `new_test` |
|---|---|---|
| 1.0 | 0.9444 | 0.9762 |
| 1.5 | 0.9583 | 0.9643 |
| 2.0 | **0.9583** | **0.9881** |
| 2.5 | 0.9444 | 0.9524 |
| 3.0 | 0.9167 | 0.9762 |
| 4.0 | 0.9583 | 0.9762 |
| 5.0 | 0.9444 | 0.9643 |

Guidance scale $s = 2.0$ achieves the best combined performance. Beyond $s = 3.0$, test accuracy drops, suggesting excessive gradient injection distorts image structure even while increasing apparent object presence.

## 4.3 EMA vs. Raw Weights

Raw model weights yield test accuracy $0.9444$ and new_test accuracy $0.9167$, compared to $0.9583$ / $0.9881$ for the EMA checkpoint. The EMA model consistently outperforms raw weights, with the gap especially pronounced on the harder `new_test` split. This

confirms that EMA's parameter averaging reduces optimisation noise and produces a smoother, more generalisable model.

## 4.4 Discussion

**Cosine vs. linear schedule.** The cosine schedule keeps $\bar{\alpha}_t$ higher for longer, ensuring the model sees a richer distribution of partially-denoised images. This leads to cleaner backgrounds and more accurate object colours at inference.

**AdaGN vs. channel concatenation.** Injecting the condition through scale/shift in every normalisation layer lets the condition modulate fine-grained spatial features. Simple concatenation at the input would propagate the condition only through network depth, which empirically gives lower accuracy on multi-label conditions.

**Guidance in $\hat{\mathbf{x}}_0$ space.** Applying the classifier gradient to the predicted clean image (rather than directly to $\mathbf{x}_t$) provides a more stable signal. At large $t$ the SNR of $\hat{\mathbf{x}}_0$ is higher than that of $\mathbf{x}_t$; clamping $\hat{\mathbf{x}}_0^*$ to $[-1, 1]$ prevents gradient blow-ups at very noisy timesteps.

# Bibliography


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